Done by: Jeffrey Harris, Aayush Ramkumar, SK Narendhran, KB Aditya

Task 1:
Create a poster that practically explains what happens in each layer of OSI model - Give examples from practical scenarios whenever feasible. → Completed.

Task 2: Explain each term in the results obtained by running "ipconfig" in your command prompt.

Connection-specific DNS Suffix: A domain name appended to single-label hostnames to resolve local domain queries.
Link-local IPv6 Address: Automatically assigned, non-routable IPv6 address used only for communication between devices on the same local network segment.
IPv4 Address: A unique 32-bit address assigned to the adapter in the local private network for local and internet routing.
Subnet Mask: It defines the portion of the IPv4 address that represents the network and which part represents the host.
Default Gateway: It is the IP address of the local router. Any packets to be sent outside the local subnet will be forwarded to this IP address to reach the internet/destination address.
Media State: Media disconnected: It indicates the network interface hardware is present but there is no active link.

Task 3: In the video of the bowling valley attack the hacker states there is no Subnetting. Explain subnetting - How it can help reduce risks.

Subnetting is the practice of dividing a single physical or logical network into two or more smaller and distinct subnetworks. Devices in different subnets cannot communicate directly at Layer 2. In the attack, subnetting would have prevented attackers from eavesdropping on unencrypted traffic and confines the initial breach to the subnet that was attacked.

Task 4: Implement a simple Java/Python code to lock/zip your folder with a password. The python script will take your folder location, take a password and encrypt the folder. → Completed.

Task 5: Create a simple login page for your website. Host it in some free platform, keep a simple password of 3 digits. Write a script to hit the URL with the passwords from 000 to 999 to see when you are getting positive response and print the password. → Completed. Simple site link: https://cs-assignment-3yo7.onrender.com/login
Github repository link: https://github.com/roxaa-aa/cs-assignment

Task 6: Implement caesar cipher to encrypt a message. Save and share the encrypted content. Other team member to read the file and decrypt the message without the key information using brute force attack. → Completed.

Task 7: Implement a simple hashing algorithm of your own. Combining the components learnt (Rotation, Shift bits, Permutation, Substitution). Ensure Determinism, Pre Image Resistance. → Completed.

Task 8: Taking one of the websites given https://www.apple.com/. Give details on each column of their certificate.

Certificate:

Version: Version 3

It indicates the version of the X.509 format used. Version 3 is the standard version that supports modular Certificate Extensions.

Serial Number: 6C:86:9E:E9:08:6B:3B:0B:15:52:C0:AB:9A:61:33:C9

It is a globally unique identifier that is assigned by the issuing Certificate Authority to distinguish this certificate from other certificates.

Certificate Signature Algorithm: PKCS #1 SHA-256 With RSA Encryption

It relays information on which algorithm is used to digitally sign the certificate to prevent tampering. This certificate uses SHA-256 and encrypts the digest with the CA's RSA private key.

Issuer: CN (Common name) = Apple Public EV Server RSA CA 1 - G1
O (Organization) = Apple Inc.
C (Country) = US

It relays information on who issued the certificate.

Validity:

Not Before: 03/07/2026, 03:41:57 GMT+5:30

Not After: 17/12/2026, 00:01:25 GMT+5:30

It identifies the exact times that this certificate is valid for.

Subject: CN (Common name) = www.apple.com
O (Organization) = Apple Inc.
L (Locality) = Cupertino
ST (State) = California
C (Country) = US
serialNumber = C0806592
jurisdictionStateOrProvinceName = California
jurisdictionCountryName = US
businessCategory = Private Organization

It gives information on which entity this certificate is granted to.

Subject Public Key Info:

Subject Public Key Algorithm: PKCS #1 RSA Encryptions

Subject Public Key: Modulus (2048 bits):

C1 80 E4 BE 25 6A 8D 8E 90 9E 26 C5 3A 94 F5 AE
BB CA 15 6C 5A CB 63 59 FC 16 97 EB 2E E3 3E 88
09 E1 42 59 EF 65 DD CA DC AC AA 63 3F 7D 76 AD
B5 F6 40 40 D5 D8 24 97 34 D9 C5 D7 8E 70 D8 84
A1 B6 49 34 D1 3A 86 A6 2B 36 F9 22 BC E6 22 50
73 AB B3 26 AD ED B3 10 62 7E FC BA 46 A8 69 C4
2F 50 F0 63 D5 FA 61 D3 15 99 E0 33 03 67 6C 4F
9F E3 F3 8D 95 5E 18 9F F4 5E 5E 19 83 B4 5E 5C
CB 2C 94 AD 74 A3 3E E4 AB 17 41 7C 46 BC 8B 9E
1D C9 CB 90 22 16 C5 6B 59 AD 66 E2 D5 52 DC 67
1C 3B 26 E3 D8 50 8D DC 77 B5 1C FF BD 11 91 5B
64 1A 77 F3 23 D7 F3 9A 61 64 73 2E D2 47 F3 05
22 72 3F AD 13 E3 C5 20 FB 81 B1 4A 21 2E 66 37
3C EE 92 DA AF 7C 29 96 6A 88 43 07 2F 01 C5 76
75 60 41 4E 95 1A 6F 47 F1 43 C6 4C E2 C7 DE 5C
9D 76 36 6D 5F F3 C4 EC 79 F1 17 81 87 47 F6 71

Public Exponent (17 bits):
01 00 01

It explains what algorithm is used for the public key. When the browser connects, it uses this key to verify TLS handshakes.

Extensions:

Certificate Basic Constraints: Critical
Is not a Certification Authority

This stops the certificate from acting as a Certificate Authority and can only be used as an end-entity server certificate which stops Apple's web server from signing certificates for other domains.

Certificate Authority Key ID: Not Critical
Key ID: D3 BD C1 3C A0 CF 35 B9 34 C5 D4 DB DA 10 0E 4C DE 6A FE 58

This is the ID of the Key of the Certificate Authority.

Authority Information Access: Not Critical
CA Issuers: URI: http://certs.apple.com/apevsrsa1g1.der
OCSP Responder: URI:
http://ocsp.apple.com/ocsp03-apevsrsa1g101

It gives information on the Authority that issued the Certificate.

Subject Certificate Alternative Name: Not Critical
DNS Name: www.apple.com.cn
DNS Name: images.apple.com
DNS Name: www.apple.com

It gives information on the alternative DNS Names that this certificate is valid for.

Certificate Policies: Not Critical
OID.2.23.140.1.1:
Certification Practice Statement Pointer:
https://www.apple.com/certificateauthority/public
OID.2.16.840.1.114412.2.1

It specifies the legal, organizational and technical rules under which the certificate was issued and how it can be used.

Extended Key Usage: Not Critical
TLS WWW Server Authentication (OID.1.3.6.1.5.5.7.3.1)

It defines the operational purposes and application protocols for which this key may be used.

CRL Distribution Points: Not Critical
URI: http://crl.apple.com/apevsrsa1g1.crl

It specifies the network locations where relying parties can download a Certificate Revocation List (CRL). A CRL is a signed file published by a CA that lists all certificates that have been revoked before their scheduled expiration date.

Certificate Subject Key ID: Not Critical
Key ID: 27 FD 01 A4 8A 09 F3 07 E5 40 7D FB 07 71 A6 78

15 58 C0 68

It is a unique identifier for the certificate subject that is derived from the public key.

Certificate Key Usage: Critical
Signing
Key Encipherment

It restricts the foundational cryptographic operations a certificate's public and private key pair is allowed to perform.

OID.1.2.840.11635.100.6.86: Not Critical
05 00

It is the custom Apple-specific Object Identifier (OID) that belongs to Apple Inc.

Signed Certificate Timestamp List: Not Critical
04 82 01 E1 01 DF 00 75 00 94 4E 43 87 FA EC C1
EF 81 F3 19 24 26 A8 18 65 01 C7 D3 5F 38 02 01
3F 72 67 7D 55 37 2E 19 D8 00 00 01 9F 24 EC AE
D0 00 00 04 03 00 46 30 44 02 20 79 0D D4 18 5F
DA DE B1 16 87 12 D3 77 B7 3A 6A FA AE CA 3A E6
A9 72 0A 44 53 35 7F 3F 25 09 66 02 20 4F FF EA
7E AE D9 1F BD 34 A8 C5 BB B6 16 16 F7 8D F7 BC
6A 91 18 9A 88 18 4E 15 73 BD 6E 76 2C 00 75 00
C8 A3 C4 7F C7 B3 AD B9 35 6B 01 3F 6A 7A 12 6D
E3 3A 4E 43 A5 C6 46 F9 97 AD 39 75 99 1D CF 9A
00 00 01 9F 24 EC AE DC 00 00 04 03 00 46 30 44
02 20 47 47 B9 62 47 5D BF 31 B2 79 00 55 5E B4
F6 2B 09 60 E2 88 56 D7 5D 59 A1 9A 5C 3D 97 00
B4 04 02 20 28 04 95 0A A5 30 00 2D 32 08 52 BF
5D 21 99 16 42 CD 5F 8A 8F 98 F2 4E C3 0B B7 1B
14 B2 81 B3 00 76 00 D7 6D 7D 10 D1 A7 F5 77 C2
C7 E9 5F D7 00 BF F9 82 C9 33 5A 65 E1 D0 B3 01
73 17 C0 C8 C5 69 77 00 00 01 9F 24 EC AF 0D 00
00 04 03 00 47 30 45 02 21 00 90 0B 6D 2A 11 37
44 E5 FB 38 DA 2C D6 2F 82 CB 68 CD AE 8B E1 B3
96 B0 F9 2D 0A DE C0 6A BE 98 02 20 1B 4D 51 90
E3 A0 4C 05 64 17 C3 33 90 76 E3 42 65 AE 7C FC
8C 8F 6B AF A6 00 AD 27 29 2C 79 7A 00 77 00 CB
38 F7 15 89 7C 84 A1 44 5F 5B C1 DD FB C9 6E F2
9A 59 CD 47 0A 69 05 85 B0 CB 14 C3 14 58 E7 00
00 01 9F 24 EC AE C6 00 00 04 03 00 48 30 46 02
21 00 FD 17 2F 9C 25 1A 51 2D 06 1B 3F 8A F2 54
03 53 05 0F 75 EE 3F 2A E4 29 A9 44 73 F6 80 58
E1 41 02 21 00 9A 93 C5 5B B3 A8 C2 2B 03 87 0D
B1 12 F3 99 01 85 30 B5 62 BF C6 42 05 F3 9B 31
48 F5 06 1E D3

It contains the cryptographic receipts that proves that a digital certificate was submitted to and recorded in the public Certificate Transparent logs before being issued.

Certificate Signing Algorithm: PKCS #1 SHA-256 With RSA Encryption
It explains the signing algorithm that was used to sign the certificate.

Certificate Signature Value: 74 67 97 17 37 F8 AE F6 9F 6E 3B 75 ED 44 7B D7
CA 5D 51 55 C6 03 CA 6F 30 95 01 F3 D9 CC C3 E8
4A F6 6A 95 B2 0D E4 11 98 24 37 3C 90 F0 69 6B

A8 E4 0F 63 7B 27 8D 79 53 46 3D 33 2A 30 47 25
41 77 C9 1F 7F 8E 1B CF 5C 3D 91 F2 94 52 93 BE
D0 8D 4A 29 6A C8 FD 89 20 50 9B 13 E6 38 FF 4E
66 23 AA FE 3F 75 F8 ED FB B1 7D 90 74 A9 E2 69
8A E0 07 5E 1C C5 0D C0 7D D2 30 52 F6 3B B9 43
75 D3 0B A2 D6 EF A9 83 5D CD 2A EC 54 B1 9F BD
6B 3E 20 2A 65 F5 27 4E 48 D0 C7 23 97 A8 A0 85
2C C7 20 E6 A6 6E 41 CA A4 45 25 72 22 67 28 6D
20 6E 4E D4 11 89 9E 8F 1C F9 A8 D6 55 72 26 92
BE 94 66 7B DD 38 13 B6 2D 7A C3 EB 60 ED 30 D0
81 BE A6 14 43 72 2E 9D 47 5D 33 FA 0A 86 EC 07
E5 6F 82 B4 0C 05 98 FC 26 6C 58 93 F0 CD 89 51
52 E6 A6 56 05 A9 FD 27 0C 75 BF B4 CF 49 E7 96

It is the actual cryptographic digital signature generated by the issuing CA.

SHA-256 Fingerprints:

Certificate: cd563f5716c254cc1a574cb9f101a923ca50b2822c94c7c85575fff4b809e760

It is the SHA-256 hash of the certificate.

Public Key: 6382171bc16bd2eb16448906aa7c37d303755bb69104d55b9e18b3bd0dad9572

It is the SHA-256 hash of the public key.

Task 9: Explain the given techniques/components and outline their use in cybersecurity:
Passkey, HSM, Key Management Service, Transparent Data Encryption, IPSec.

Passkey: It is an authentication standard which uses public-key asymmetric cryptography. The user's device generates a unique key pair, where the public key is stored on the authentication servers and the private key is secured on the user's local device. Its use in cybersecurity: It allows for phishing resistance as keys are fixed to a specific URL. Thus, faking/spoofing the URL won't matter as the browser won't allow the connection to complete.
HSM: It is a dedicated physical, tamper-responsive computing device that is used specifically to generate, store and manage cryptographic keys. They wipe their internal data if any tampering occurs. Its use in cybersecurity: As the cryptographic operations occur within the HSM's secure location, private and secret keys are never leaving the hardware boundary in plaintext.
Key Management Service: It is a centralised software/cloud-managed service that governs the entire lifecycle of a cryptographic key. It governs the generation, distribution, rotation, access control, auditing and deletion of the keys. Its use in cryptography: It's used for access governance and least privilege. It binds access to Identity and Access Management policies which allows only authorised users to invoke keys to decrypt data.
Transparent Data Encryption: It's a database-level technology that automatically encrypts files at rest without requiring any changes to the existing database queries or code. Its use in cybersecurity: It protects against theft, storage loss or unauthorized copying. Even if the malicious actor gains access to the database files they will not be able to read it. It also allows organizations to meet regulatory requirements of HIPPA and GDPR without changing the application logic.
IPSec: It is a suite of open-standard cryptographic protocols that operate at Layer 3, the network layer of the OSI model. Its use in cybersecurity: It allows for secure network tunnels that are used in VPNs. It protects traffic across untrusted public networks that are used for site-to-site VPNs and client-to-site remote access.

Task 10: Add a symmetric key to your Task 7 → Convert it from Hashing Algorithm to MAC. → Completed.